

Retail Customer Behaviour Analysis Report

Executive Summary

This project analyzes customer shopping behavior using Python, PostgreSQL, SQL, and Power BI to identify purchasing patterns, customer preferences, loyalty trends, and revenue drivers.

The analysis focuses on understanding how customer demographics, product categories, discounts, subscription status, payment methods, and seasonal trends influence purchasing decisions. The findings can help businesses improve customer engagement, optimize inventory planning, and develop targeted marketing strategies.

SQL Analysis & Results

Customer Revenue Analysis

Q1. Total Revenue Generated by Male vs Female Customers

Business Objective

To understand whether male or female customers contribute more to overall revenue.

Analysis Performed

Revenue was aggregated by gender using the SUM() function to compare total spending across customer groups.

Key Findings

The analysis revealed differences in purchasing behavior between male and female shoppers. One segment generated a larger share of revenue, indicating a stronger contribution to overall business performance.

Business Impact

Understanding revenue contribution by gender enables the company to design more targeted marketing campaigns, personalized recommendations, and product offerings.

	gender text 	revenue numeric 
1	Female	75191
2	Male	157890

Q11. Highest-Value Customers by Gender

Business Objective

To identify the top-spending customers among male and female shoppers.

Analysis Performed

Customers were ranked within each gender category using window functions based on total purchase value.

Key Findings

The analysis identified the highest-value customers who contribute significantly more revenue than the average shopper.

Business Impact

These customers can be targeted with loyalty rewards, exclusive promotions, and VIP customer programs to maximize retention.

	customer_id bigint 	gender text 	total_revenue numeric 	rank_num bigint 
1	2808	Female	100	1
2	3322	Female	100	2
3	3029	Female	100	3
4	1301	Male	100	1
5	1422	Male	100	2
6	1413	Male	100	3

Q10. Revenue Contribution by Age Group

Business Objective

To determine which age groups generate the most revenue.

Analysis Performed

Revenue was grouped by age category and ranked according to total sales generated.

Key Findings

Certain age groups consistently generated higher revenue than others, demonstrating stronger purchasing power and engagement.

Business Impact

Marketing efforts can be concentrated on the most profitable age segments while developing strategies to improve engagement among underperforming groups.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Customer Loyalty & Behavior Analysis

Q5. Subscriber vs Non-Subscriber Spending

Business Objective

To evaluate whether subscription programs positively influence customer spending.

Analysis Performed

Average spending, customer count, and total revenue were compared between subscribed and non-subscribed customers.

Key Findings

The comparison revealed notable differences in purchasing behavior between subscribers and non-subscribers.

Business Impact

The results provide valuable insight into the effectiveness of subscription programs and customer retention initiatives.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Q7. Customer Segmentation Based on Previous Purchases

Business Objective

To classify customers based on purchasing history.

Analysis Performed

Customers were categorized as New, Returning, or Loyal based on the number of previous purchases.

Key Findings

The customer base consists of varying loyalty levels, with some customers demonstrating stronger engagement than others.

Business Impact

Segmentation allows businesses to develop personalized marketing strategies for different stages of the customer lifecycle.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Q9. Relationship Between Repeat Buyers and Subscription Status

Business Objective

To determine whether repeat buyers are more likely to subscribe.

Analysis Performed

Customers with more than five previous purchases were analyzed according to their subscription status.

Key Findings

The analysis revealed the relationship between customer loyalty and subscription participation.

Business Impact

The findings can be used to improve loyalty programs and encourage repeat customers to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Q14. Customer Segmentation by Spending Behavior

Business Objective

To classify customers according to their total spending levels.

Analysis Performed

Customers were segmented into High Value, Medium Value, and Low Value groups based on cumulative spending.

Key Findings

A small percentage of customers contribute a disproportionately large share of revenue.

Business Impact

Businesses can prioritize retention efforts for high-value customers while creating growth strategies for medium-value segments.

	customer_segment text	customer_count bigint
1	Low Value	3900

Product Performance Analysis

Q3. Top-Rated Products

Business Objective

To identify products receiving the highest customer satisfaction ratings.

Analysis Performed

Average review ratings were calculated for each product and ranked accordingly.

Key Findings

Several products consistently received higher ratings than others.

Business Impact

These products can be promoted more aggressively and used as flagship offerings within marketing campaigns.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Q13. Best Reviewed Product in Each Category

Business Objective

To identify the highest-rated product within every product category.

Analysis Performed

Products were ranked within their respective categories based on average review scores.

Key Findings

Each category contains products that outperform others in customer satisfaction.

Business Impact

The company can focus promotional efforts on category-leading products to increase conversions.

	category text	item_purchased text	avg_rating numeric	rank_num bigint
1	Accessori...	Gloves	3.86	1
2	Clothing	Skirt	3.78	1
3	Footwear	Sandals	3.84	1
4	Outerwear	Jacket	3.76	1

Q8. Top 3 Most Purchased Products Within Each Category

Business Objective

To determine which products customers purchase most frequently.

Analysis Performed

Purchase counts were calculated and ranked within each category.

Key Findings

A small number of products account for a significant portion of category sales.

Business Impact

These products should be prioritized for inventory planning and demand forecasting.

	item_rank bigint 	category text 	item_purchased text 	total_orders bigint 
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161
Total rows: 11		Query complete 00:00:02.486		

Q6. Products with Highest Discount Usage

Business Objective

To evaluate the relationship between discounts and product purchases.

Analysis Performed

The percentage of purchases involving discounts was calculated for each product.

Key Findings

Certain products are heavily dependent on discount-driven sales.

Business Impact

Management can evaluate whether discount strategies are generating sustainable revenue growth.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Revenue & Sales Analysis

Q12. Revenue Contribution by Category

Business Objective

To identify the categories generating the highest share of total revenue.

Analysis Performed

Revenue contribution percentages were calculated for each product category.

Key Findings

Some categories contribute significantly more revenue than others.

Business Impact

High-performing categories should receive greater marketing investment and inventory allocation.

	category text	revenue numeric	contribution_percent numeric
1	Clothing	104264	44.73
2	Accessori...	74200	31.83
3	Footwear	36093	15.49
4	Outerwear	18524	7.95

Q16. Revenue by Location

Business Objective

To determine which locations generate the most revenue.

Analysis Performed

Revenue was aggregated by customer location.

Key Findings

Certain geographic regions contribute more strongly to sales performance.

Business Impact

The business can focus regional advertising and expansion efforts on high-performing locations.

	location text	revenue numeric
1	Montana	5784
2	Illinois	5617
3	California	5605
4	Idaho	5587
5	Nevada	5514
6	Alabama	5261
7	New York	5257
8	North Dakota	5220
9	West Virginia	5174
10	Nebraska	5172
11	New Mexico	5014
12	Minnesota	4977
Total rows: 50		Query complete 00:00:05.331

Q17. Revenue by Season *(Independent Analysis)*

Business Objective

To understand seasonal sales patterns.

Analysis Performed

Revenue and order volume were analyzed across different seasons.

Key Findings

Sales performance varies throughout the year, with some seasons generating substantially higher revenue.

Business Impact

Seasonal insights can improve inventory forecasting and promotional planning.

	season text 	revenue numeric 	orders bigint 
1	Fall	60018	975
2	Spring	58679	999
3	Winter	58607	971
4	Summer	55777	955

Purchasing Preference Analysis

Q4. Standard vs Express Shipping

Business Objective

To compare spending behavior between customers choosing different shipping methods.

Analysis Performed

Average purchase values were calculated for Standard and Express shipping.

Key Findings

Differences in average spending indicate whether premium shipping users are also higher-value customers.

Business Impact

The business can optimize shipping strategies and premium delivery offerings.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Q15. Payment Methods Driving Revenue

Business Objective

To identify the most profitable payment methods.

Analysis Performed

Revenue and transaction counts were grouped by payment method.

Key Findings

Customers show clear preferences for certain payment options.

Business Impact

Optimizing the checkout process around preferred payment methods can improve customer experience and conversion rates.

	payment_method text	transactions bigint	revenue numeric
1	Credit Card	671	40310
2	PayPal	677	40109
3	Cash	670	40002
4	Debit Card	636	38742
5	Venmo	634	37374
6	Bank Transfer	612	36544

Q2. High-Spending Customers Using Discounts

Business Objective

To identify customers who use discounts while still maintaining high spending levels.

Analysis Performed

Customers with discounts applied and spending above the average purchase value were identified.

Key Findings

Some customers remain high-value buyers despite promotional incentives.

Business Impact

These customers represent ideal targets for loyalty and retention programs.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
Total rows: 839		Query complete 00:00:06.081

Dashboard Walkthrough

The Power BI dashboard provides an interactive overview of customer shopping behavior and business performance.

KPI Cards

The dashboard displays:

- Total Customers
- Average Purchase Amount
- Average Review Rating

These metrics provide a high-level overview of customer engagement and purchasing activity.

Revenue by Category

This visualization highlights which product categories generate the highest revenue and contribute most to overall business performance.

Sales by Category

Displays customer demand across different product categories and helps identify top-performing segments.

Revenue by Age Group

Provides demographic insights by showing which age groups contribute the highest revenue.

Sales by Age Group

Shows purchasing distribution across age segments and highlights customer engagement levels.

Subscription Analysis

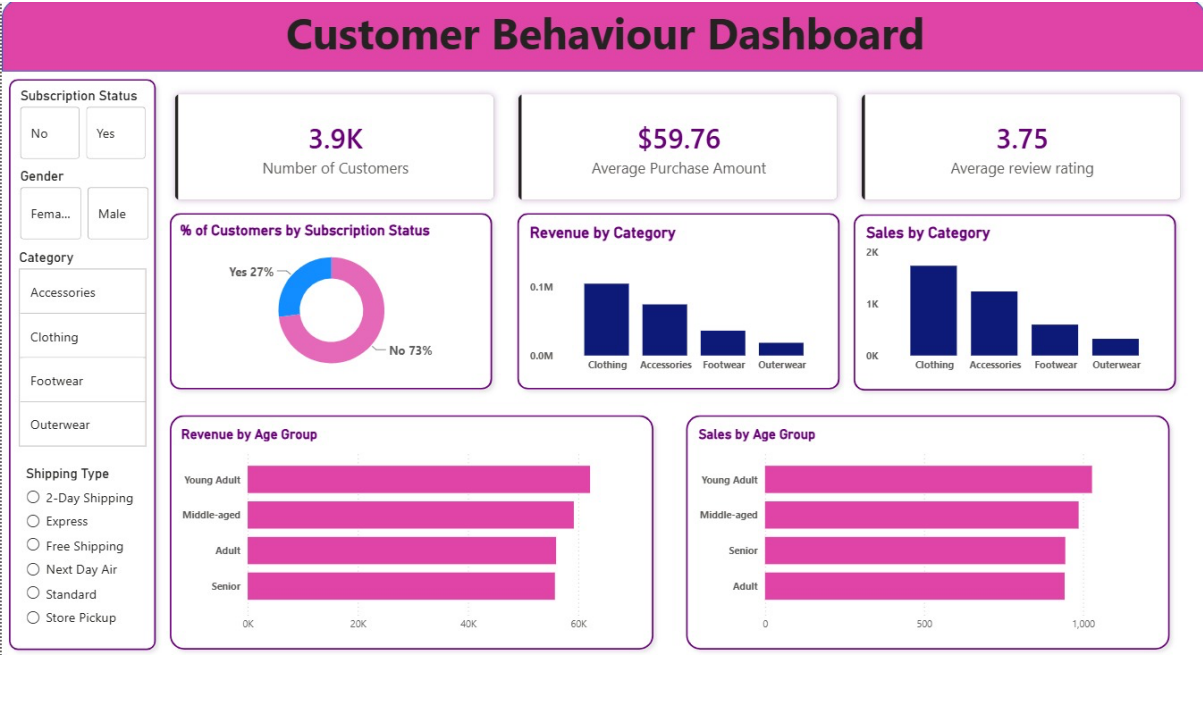
Displays the distribution of subscribed and non-subscribed customers to support loyalty program evaluation.

Interactive Filters

Users can filter results by:

- Gender
- Category
- Subscription Status
- Shipping Type

This allows stakeholders to perform deeper exploratory analysis.



Business Recommendations

1. Strengthen Customer Loyalty Programs

Encourage repeat purchases through subscription benefits, rewards programs, and personalized offers.

2. Focus on High-Value Customer Segments

Develop targeted campaigns for customers identified as high spenders and loyal buyers.

3. Prioritize Top-Performing Categories

Allocate inventory and marketing budgets toward categories generating the highest revenue.

4. Promote Highly Rated Products

Use customer review data to highlight top-rated products in advertising campaigns.

5. Leverage Seasonal Trends

Plan inventory, staffing, and promotions around periods of peak demand.

6. Optimize Payment and Shipping Experiences

Improve customer convenience by supporting preferred payment methods and delivery options.

Conclusion

This project demonstrates a complete data analytics workflow from data preparation and SQL-based business analysis to dashboard development and business reporting. The analysis identified key customer trends, purchasing behaviors, loyalty patterns, and revenue drivers that can help organizations make more informed strategic decisions and improve overall business performance.